

F78PB Probability and Statistics B

Confidence Intervals

Confidence interval estimation

We will consider confidence interval (CI) estimation under the *classical inference* approach.

Let $X_1, X_2, \ldots, X_n$ denote a *random sample* from a distribution specified by a pdf (or pmf if discrete) $f_X(x;\theta)$ where θ is an *unknown parameter*.

We have considered methods for obtaining a point estimate of θ from the random sample $\underline{X}$.

Now we consider interval estimation as a way of specifying *accuracy*.

Aim: Find a lower limit $\theta_L(\underline{X})$ and upper limit $\theta_U(\underline{X})$ such that

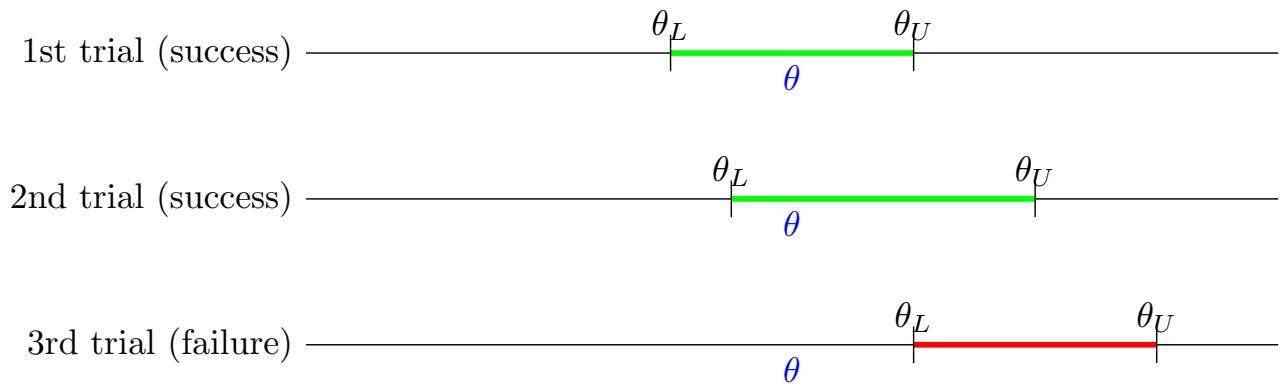
$$\mathbb{P}\{\theta_L(\underline{X}) < \theta < \theta_U(\underline{X})\} = 1 - \alpha$$

for $0 < \alpha < 1$.

The interval $(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is called a $100(1-\alpha)\%$ confidence interval for θ.

Important remarks:

- θ is **not** random
- But $\theta_L(\underline{X})$ and $\theta_U(\underline{X})$ **are** random
- This means that $(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is a random interval.
  Each time we repeat the 'experiment' we will obtain a different interval!



$(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is a $100(1-\alpha)\%$ CI $\Leftrightarrow \mathbb{P}(\theta \in (\theta_L(\underline{X}), \theta_U(\underline{X}))) = (1-\alpha)$.

Example

Construct a 95% confidence interval for population mean μ (assume σ^2 is known) when the sample size n is large. Note that we don't make any assumptions about the distribution of the population.

Solution:

We know that $\bar{X}$ is a good point estimator for μ. To obtain a CI for μ we need information about the distribution of $\bar{X}$.

Remember, since n is large the central limit theorem gives

$$\bar{X} \dot{\sim} \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \Rightarrow \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \dot{\sim} \mathrm{N}(0,1)$$

$$\Rightarrow \mathbb{P}\left(-1.96 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1.96\right) \approx 0.95 \quad \text{(Tables)}$$

Rearranging the inequality we obtain

$$\mathbb{P}\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

So for large n and σ^2 known

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is *approximately* a 95% CI for μ.

Notice that this is of the form $(\theta_L(\underline{X}), \theta_U(\underline{X}))$.

Pivotal quantities

The derivation of the CI in the previous example worked because we used a pivotal quantity.

$h(\underline{X}, \theta)$ is a pivotal quantity for θ if:

1. $h(\underline{X}, \theta)$ is a function of both the sample $\underline{X}$ and the parameter θ.
   - For our example, $h(\underline{X}, \mu) = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is a function of $\underline{X}$ and μ.
2. $h(\underline{X}, \theta)$ has a known (or *approximately* known) distribution that does not depend on θ.
   - We use this knowledge to find constants a and b such that
   $$\mathbb{P}(a < h(\underline{X}, \theta) < b) = 1 - \alpha.$$
   - In our example, $h(\underline{X}, \mu) \dot{\sim} \mathrm{N}(0, 1)$, so we know from properties of the normal distribution that
   $$\mathbb{P}\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) \approx 0.95$$
3. $h(\underline{X}, \theta)$ is monotonic (often linear) function of θ.
   - So we can invert the inequality $a < h(\underline{X}, \theta) < b$ to produce an inequality with θ 'in the middle'.

Further observations based on the Example

The interval $\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$ is **random**.

Before we observe the data, the probability is 95% that the interval will contain the unknown μ.

After observing the data, we have a value $\bar{x}$ and a specific interval $\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$.

This either contains μ or doesn't contain μ (and we are not certain about this!).

Nevertheless, we can say that we are 95% confident that the interval contains μ. (If we repeat the process for a large number of times, on 95% of occasions the interval will contain μ.)

The CI derived above is a 95% CI for μ.

In general we can obtain a $(1-\alpha)100\%$ CI for μ using $z_{\alpha/2}$ instead of 1.96, i.e.

$$\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

For example we get:

- a 90% CI for μ with $z_{0.05} = 1.645$
- a 99% CI for μ with $z_{0.005} = 2.576$
- a 50% CI for μ with $z_{0.25} = 0.675$

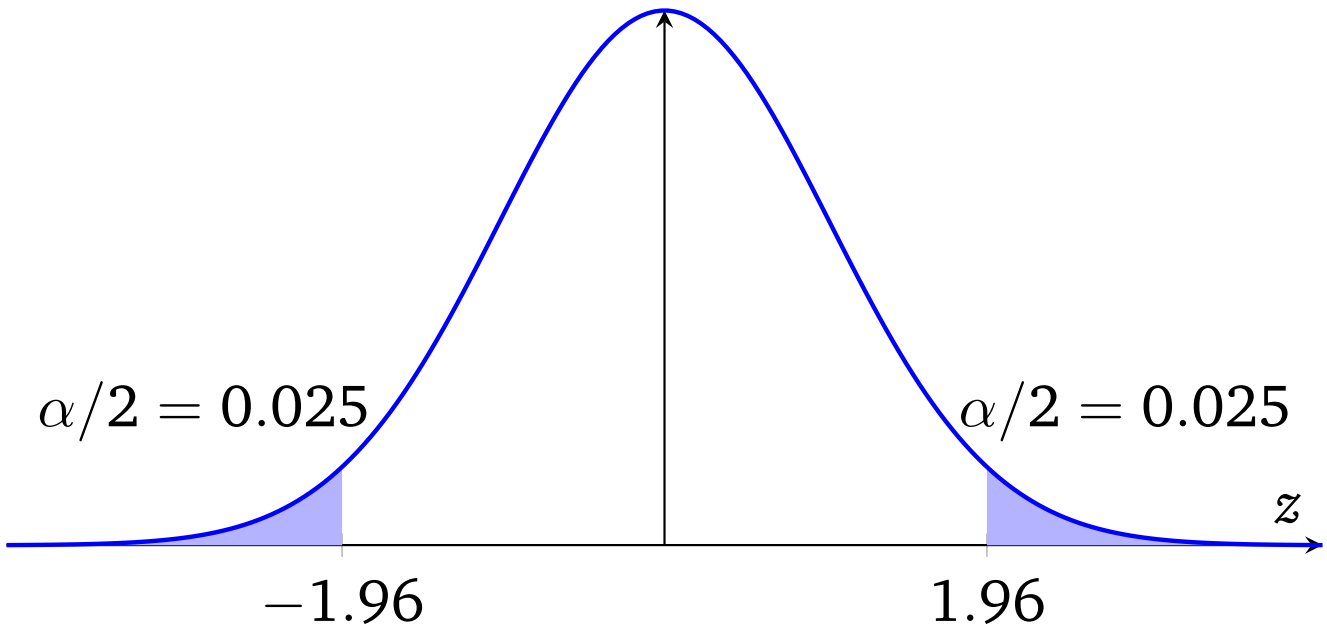
Confidence Intervals: Equal-Tailed

- In our example, we derived an **equal-tailed** 95% CI:

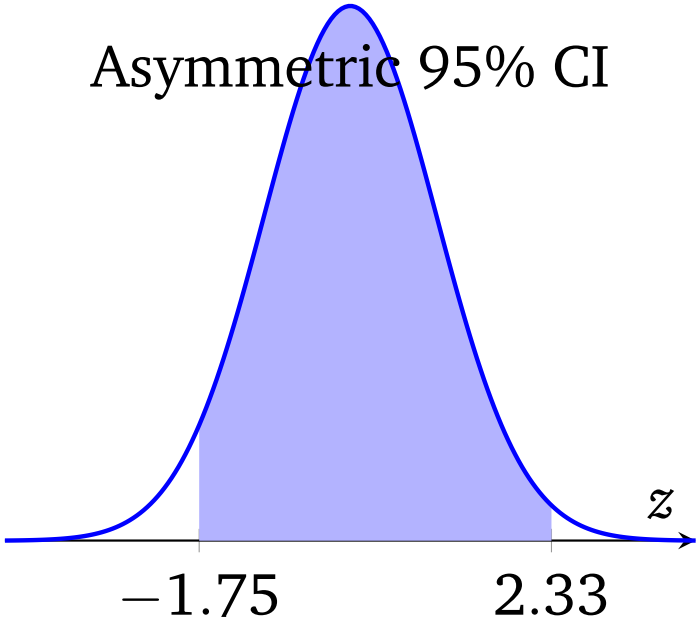
$$\mathbb{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < -1.96\right) = \mathbb{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > 1.96\right) \approx 0.025$$

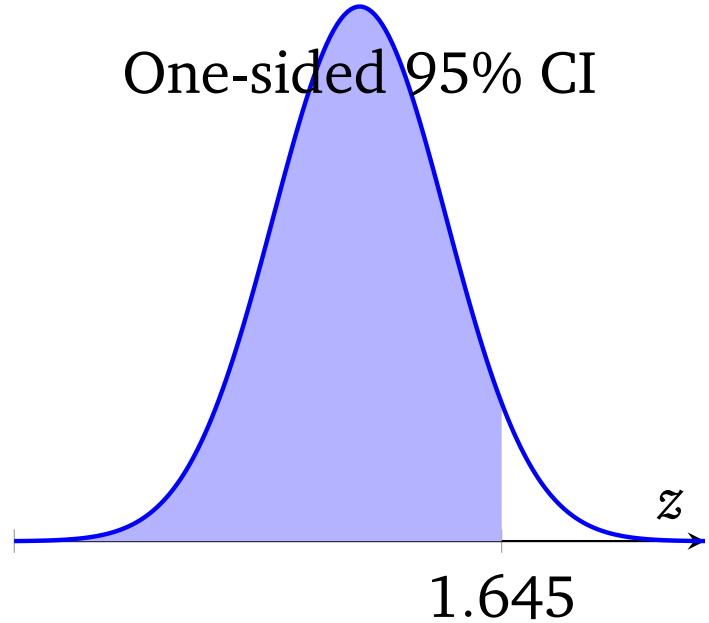
- This gives a 95% confidence interval for μ as

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$



Other Confidence Intervals





$$\mathbb{P}\left(-1.75 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 2.33\right) \approx 0.95$$
$$\Rightarrow \left(\bar{X} - 2.33\frac{\sigma}{\sqrt{n}},\ \bar{X} + 1.75\frac{\sigma}{\sqrt{n}}\right)$$

$$\mathbb{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1.645\right) \approx 0.95$$
$$\Rightarrow \left(-\infty,\ \bar{X} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$

Here it seems 'natural' to construct an equal-tailed CI since the *sampling distribution* of $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is (approximately) symmetric. This implies that the equal-tailed CI is the narrower 95% confidence interval.

So far we have a method for constructing a CI for μ based on

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} \mathrm{N}(0,1) \quad \text{for } \textbf{large } n \text{ and } \textbf{known } \sigma^2$$

(no distributional assumptions).

How can we construct CIs for μ (or other parameters) when:

- n is not large?
- σ is not known?
- population has a particular distribution?

Strategy: Try to find a *pivotal quantity* based on the MLE of the parameter and then find its *sampling distribution*.

Sampling from a normal $\mathrm{N}\left(\mu, \sigma^2\right)$ population

Suppose $X_1, X_2, \ldots, X_n$ is a random sample from a $\mathrm{N}\left(\mu, \sigma^2\right)$ distribution with both μ and σ^2 *unknown*.

Remember:

- MLE for μ is $\hat{\mu}(\underline{X}) = \bar{X}$
- MLE for σ^2 is $\hat{\sigma}^2(\underline{X}) = \frac{1}{n}\sum_i (X_i - \bar{X})^2 = \frac{n-1}{n} S^2$

Sampling distributions for $\bar{X}$ and S^2

1. $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
2. $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
3. $\bar{X}$ and S^2 are **independent**

Chi-square with n degrees of freedom (χ^2_n)

1. χ^2_n is a particular type of Gamma distribution:

$$\chi^2_n = \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right).$$

   Note: n is a *parameter* in the density for χ^2_n.
   Calculations involving χ^2_n random variables are done using statistical tables.

2. χ^2_n has the **same distribution** as the sum of n i.i.d. χ^2_1 random variables (since the sum of i.i.d. Gamma r.v. is another Gamma r.v.).

3. If $Z \sim \text{N}(0,1)$, then $Z^2 \sim \chi^2_1$. It follows from this and from (2) that if $Z_1, Z_2, ..., Z_n$ are i.i.d. $\text{N}(0,1)$ random variables, then

$$Z_1^2 + Z_2^2 + \cdots + Z_n^2 \sim \chi^2_n.$$

4. **Key fact:** If the populations distribution is $\text{N}\left(\mu, \sigma^2\right)$ then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

t-distribution with n degrees of freedom (t_n)

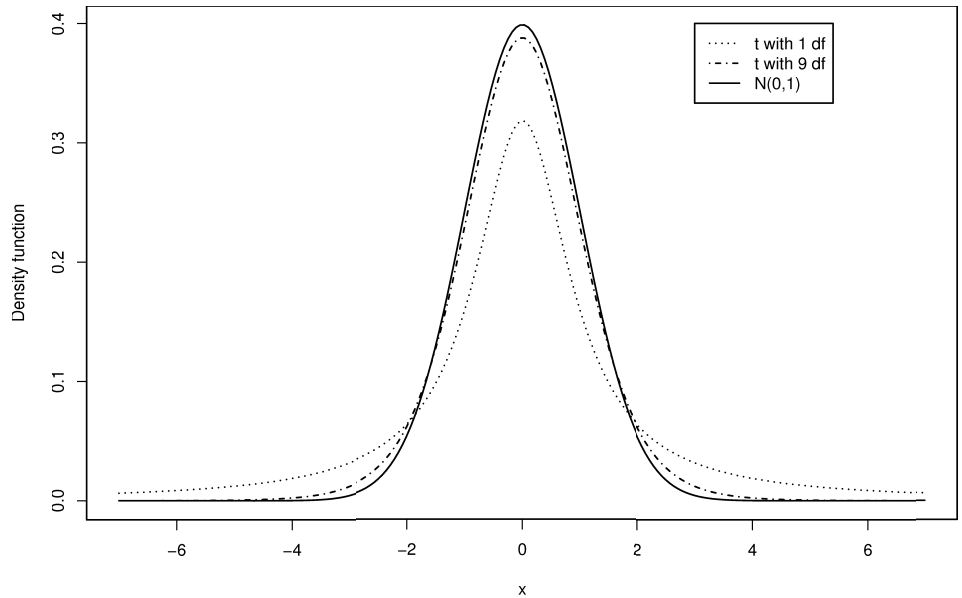
The t distribution (or *Student's t* – named after the pseudonym 'Student' that W.S. Gosset used) is a continuous distribution.

The density for the t distribution with n degrees of freedom is given by

$$f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n}\Gamma(\frac{n}{2})}\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}} \qquad \text{for} \quad -\infty < x < \infty.$$

t-distribution with n degrees of freedom (t_n)

Plot of pdf:



Note that:

1. The t distribution is symmetric around zero, with longer tails than the $\mathrm{N}(0, 1)$.
2. As $n \to \infty$, the t distribution approaches the $\mathrm{N}(0, 1)$ distribution.
3. The values of its cumulative distribution function are tabulated (e.g. see *NCST* p42–45).

Recipe for obtaining a random variable with the t_n distribution:

If $U \sim \mathrm{N}(0,1)$ and $V \sim \chi^2_n$ and U and V are **independent**, then the variable

$$T = \frac{U}{\sqrt{V/n}}$$

has a t distribution with n degrees of freedom.

Key Fact: Suppose that $\underline{X}$ is a random sample from $\mathrm{N}\left(\mu, \sigma^2\right)$, then

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Proof.

We know that $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ and that these statistics are **independent**. The above recipe implies that

$$\frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{S^2}{\sigma^2}}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

F distribution with (n_1, n_2) degrees of freedom (F_{n_1,n_2})

(1) An F random variable is constructed from the quotient of two χ^2 random variables.
In particular, suppose that $U \sim \chi^2_{n_1}$ and $V \sim \chi^2_{n_2}$ and that U and V are **independent**, then the distribution of the random variable

$$F = \frac{U/n_1}{V/n_2}$$

is called an F distribution with (n_1, n_2) degrees of freedom.

(2) An F random variable is always positive and the shape of the density is similar to the shape of a χ^2 density.
Calculations involving the F distribution are done using statistical tables.

Confidence intervals for means for μ - normal case

(1) Population variance σ^2 known

Let $X_1, X_2, \ldots, X_n$ denote a random sample from $\mathrm{N}\left(\mu, \sigma^2\right)$. When the population variance σ^2 is *known*, the pivotal quantity $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$ (exactly).

A $(1-\alpha)100\%$ equal-tailed CI for μ is given by

$$\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $z_{\frac{\alpha}{2}}$ is the appropriate percentage point of the $\mathrm{N}(0,1)$ distribution, i.e. it is such that $\mathbb{P}\left(Z > z_{\alpha/2}\right) = 1 - \Phi\left(z_{\alpha/2}\right) = \frac{\alpha}{2}$.

(2) Population variance σ^2 not known

We can use the pivotal quantity $h(\underline{X}, \mu) = \frac{\bar{X}-\mu}{S/\sqrt{n}}$. We have

$$h(\underline{X}, \mu) = \frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{S^2}{\sigma^2}}}$$

with the numerator following $\mathrm{N}(0,1)$ and $(n-1)S^2/\sigma^2 \sim \frac{\chi^2_{n-1}}{n-1}$. Also, numerator and denominator are *independent*.

It follows that $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$ (and is also *monotonic* in μ).

Then, a $(1-\alpha)100\%$ equal-tailed CI for μ will be given by

$$\left(\bar{X} - t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}}\right)$$

where $t_{n-1,\frac{\alpha}{2}}$ is the point of the t_{n-1} distribution such that $\mathbb{P}(T > t_{n-1,\frac{\alpha}{2}}) = \frac{\alpha}{2}$, with $T \sim t_{n-1}$.

A $(1-\alpha)100\%$ equal-tailed CI for μ will be given by

$$\left(\bar{X} - t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}}\right)$$

where $t_{n-1,\frac{\alpha}{2}}$ is the point of the t_{n-1} distribution such that $\mathbb{P}(T > t_{n-1,\frac{\alpha}{2}}) = \frac{\alpha}{2}$, with $T \sim t_{n-1}$.

For example, 95% equal-tailed CI for μ is given by

$$\left(\bar{X} - t_{n-1,0.025}\frac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,0.025}\frac{S}{\sqrt{n}}\right) \quad (\textit{use Tables})$$

For example, 95% equal-tailed CI for μ is given by

$$\left(\bar{X} - t_{n-1,0.025}\frac{S}{\sqrt{n}},\ \ \bar{X} + t_{n-1,0.025}\frac{S}{\sqrt{n}}\right) \quad (\textit{use Tables})$$

Remarks

1. This result is *exact* for normal populations. It also gives an approximate CI for non-normal populations.
2. For large n, the t_n distribution can be approximated by the $\mathrm{N}(0,1)$ distribution (e.g. $t_{n-1,0.025} \approx 1.96$).

Example

The population distribution of the length (measured in microns) of a certain kind of bacteria is considered to be Normal with *known* variance 0.25 and unknown mean μ. Eight bacteria are selected at random and their lengths measured. The resulting data are:

$$6.3 \quad 7.3 \quad 6.6 \quad 6.8 \quad 8.0 \quad 7.6 \quad 7.1 \quad 7.0$$

(a) Calculate a 90% confidence interval for the population mean μ.

(b) Suppose now that the population variance is *unknown*. Calculate a 90% confidence interval for μ in this case.

Solution ...

Confidence intervals for σ^2

We use $\frac{(n-1)S^2}{\sigma^2}$ as a pivotal quantity for σ^2 since:

- it is a function of $\underline{X}$ and σ^2
- it has a *known* distribution (χ^2_{n-1}) which does not depend on σ^2
- it is monotonic in the parameter σ^2.

Use the % tables of the χ^2 distribution (e.g. NCST p40, 41) to find values $\chi^2_{n-1,1-\frac{\alpha}{2}}$ and $\chi^2_{n-1,\frac{\alpha}{2}}$ such that

$$\mathbb{P}\left(Y > \chi^2_{n-1,\frac{\alpha}{2}}\right) = \frac{\alpha}{2}, \quad \text{with } Y \sim \chi^2_{n-1}.$$

giving

$$\mathbb{P}\left(\chi^2_{n-1,1-\frac{\alpha}{2}} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{n-1,\frac{\alpha}{2}}\right) = 1-\alpha$$

Rearrange inequality to obtain

$$\mathbb{P}\left(\frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}\right) = 1-\alpha$$

So, for example, a 95% equal-tailed CI for σ^2 is given by

$$\left(\frac{(n-1)S^2}{\chi^2_{n-1,0.025}}, \quad \frac{(n-1)S^2}{\chi^2_{n-1,0.975}} \right)$$

Note that this is **not** the narrowest 95% CI, but it is *easy* to compute and *standard*.

Example

Find a 95% CI for σ^2 for a sample of size $n = 12$ (from a normal population) that has sample variance 32.93.

Solution ...

Example

Now find a 90% CI for σ^2 for a sample of size $n = 25$ (from a normal population) where the sample variance 10.

Solution ...

Approximation for large n

Tables only give percentage points for $n \leq 100$.

What do we use for $n > 100$?

Recall:

if $Z_1, Z_2, \ldots, Z_n$ are i.i.d. $\mathrm{N}(0,1)$ random variables,

then $Z = Z_1^2 + Z_2^2 + \ldots + Z_n^2 \sim \chi^2_n$, with $\mathbb{E}(Z) = n$ and $\mathbb{V}\mathrm{ar}(Z) = 2n$.

Then Central Limit Theorem gives

$$\frac{Z - n}{\sqrt{2n}} \dot{\sim} \mathrm{N}(0,1).$$

Application:

$$\frac{\frac{(n-1)S^2}{\sigma^2} - (n-1)}{\sqrt{2(n-1)}} \dot{\sim} \mathrm{N}(0,1).$$

Use this as an approximate pivotal quantity.

Comparing two random variables

We have two random variables X and Y with means μ_X, μ_Y and variances σ_X^2, σ_Y^2 respectively and we would like to compare these random variables.

- What can we say about the difference $\mu_X - \mu_Y$?
- What can we say about the ratio σ_X^2/σ_Y^2?

We will assume that we have two **independent** samples of size n_X and n_Y:

$$\underline{X} = (X_1, \ldots, X_{n_X}),$$
$$\underline{Y} = (Y_1, \ldots, Y_{n_Y}).$$

Comparison of means

How should we *estimate* the difference $\mu_X - \mu_Y$?

We know that

$$\bar{X} = \frac{X_1 + X_2 + \ldots + X_{n_X}}{n_X}$$

is an unbiased estimator for μ_X, and likewise

$$\bar{Y} = \frac{Y_1 + Y_2 + \ldots + Y_{n_Y}}{n_Y}$$

is an unbiased estimator for μ_Y, so $\bar{X} - \bar{Y}$ is an unbiased estimator for $\mu_X - \mu_Y$.

Can we develop a confidence interval for the difference $\mu_X - \mu_Y$?

Case 1: (large sample sizes)

Suppose that the sample sizes n_1 and n_2 are *large*. The *Central Limit Theorem* tells us that

$$\bar{X} \dot{\sim} \mathrm{N}\left(\mu_X, \frac{\sigma_X^2}{n_X}\right) \quad \text{and} \quad \bar{Y} \dot{\sim} \mathrm{N}\left(\mu_Y, \frac{\sigma_Y^2}{n_Y}\right).$$

Since the samples are independent, the sample means $\bar{X}$ and $\bar{Y}$ are **independent** too, and the sum of independent (approximately) normal random variable is also (approximately) normal. So we have

$$\bar{X} - \bar{Y} \dot{\sim} \mathrm{N}\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}\right).$$

It follows that

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \dot{\sim} \mathrm{N}(0, 1)$$

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \dot{\sim} \mathrm{N}(0, 1)$$

We can use this to develop the following $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$:

$$\left(\bar{X} - \bar{Y} - z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}, \bar{X} - \bar{Y} + z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \right).$$

Remark

In all above results, if σ_1^2, σ_2^2 are *not known* we can replace them with S_1^2, S_2^2 for an approximate CI.

Case 2: Normal populations with known variances

Assumptions:

$$X \sim \mathrm{N}\left(\mu_X, \sigma_X^2\right), \quad Y \sim \mathrm{N}\left(\mu_Y, \sigma_Y^2\right).$$

Then the sample means have exact distributions:

$$\bar{X} \sim \mathrm{N}\left(\mu_X, \frac{\sigma_X^2}{n_X}\right), \qquad \bar{Y} \sim \mathrm{N}\left(\mu_Y, \frac{\sigma_Y^2}{n_Y}\right).$$

The standardized difference is:

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \sim \mathrm{N}(0, 1).$$

If σ_X^2 and σ_Y^2 are **known**, then Z is a pivotal quantity. The 95% confidence interval for $(\mu_X - \mu_Y)$ is:

$$\left((\bar{X} - \bar{Y}) - 1.96\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}},\ (\bar{X} - \bar{Y}) + 1.96\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \right).$$

Case 3: Normal populations with unknown variances

Assumptions:

$$\sigma_X^2 \text{ and } \sigma_Y^2 \text{ are unknown, but } \sigma_X^2 = \sigma_Y^2 = \sigma^2.$$

From earlier results:

$$\frac{(n_X - 1)S_X^2}{\sigma^2} \sim \chi^2_{n_X - 1}, \qquad \frac{(n_Y - 1)S_Y^2}{\sigma^2} \sim \chi^2_{n_Y - 1}.$$

These are independent (since samples are independent). Therefore:

$$\frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{\sigma^2} \sim \chi^2_{n_X + n_Y - 2}.$$

Define the **pooled variance**:

$$S_p^2 = \frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_X + n_Y - 2}.$$

Using independence and previous results, the pivotal quantity is:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1 + n_2 - 2}.$$

Using this pivotal quantity we obtain the $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ as:

$$\left(\bar{X} - \bar{Y} - t_{n_1+n_2-2,\alpha/2} \sqrt{S_p^2 \left(\frac{1}{n_X} + \frac{1}{n_Y} \right)}, \right.$$
$$\left. \bar{X} - \bar{Y} + t_{n_1+n_2-2,\alpha/2} \sqrt{S_p^2 \left(\frac{1}{n_X} + \frac{1}{n_Y} \right)} \right).$$

Remarks

- The validity of this construction depends on the assumption that the populations are normally distributed and that the variances are equal.
- You should check these assumptions by looking at *plots* of the data *before* applying the result.
- If the plots appear to be 'bell-shaped' with similar spread, then the procedure outlined above will yield a reasonable CI even if the populations are only approximately normal.
- However, if the variances (i.e. the spread of the data) appear to be significantly different, this approach should not be used.